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Options and Futures

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Introduction to Fixed Income and Linear Interest Rate Derivatives

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Outline

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- Introduction to basic fixed-income concepts:
  - Zero-coupon bonds, yield curve, bond pricing, forward interest rates
- Forward rate agreements (FRAs)
- Interest rate swaps

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Introduction to basic fixed-income concepts

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- So far, for the most part, we had assumed a constant interest rate r.
  - We were using the discount factor $e^{-r(T-t)}$ to evaluate, as of time t, a riskfree cash flow paid at time T
- In reality there are many different interest rates, e.g. for different maturities (the yield curve) or for different credit ratings (credit spreads).
  - We will ignore credit /counterparty risk.
- Moreover these interest rates move over time in a stochastic fashion. This generates "interest rate risk" that many market participants are exposed to.
  - This type of risk can be traded or hedged using interest rate derivatives.

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Zero-coupon bonds and zero rates

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- Consider a zero-coupon bond (also referred to simply as a "zero-coupon" or "zero") with face value \$1 maturing at date T.
  - One deterministic cash flow: The bond pays \$1 at time T
- Let $P(t, T)$ denote the price of that bond at time t with $t \leq T$.
  - Absence of arbitrage requires that $P(T, T) = 1$. In addition, assuming interest rates are always positive, it must be that $P(t, T)$ is decreasing in T.
- The yield-to-maturity of the zero-coupon at time t, $y(t, T)$, defined as a continuously compounded rate, is such that:

$$P(t, T) = e^{-y(t,T)(T-t)}$$

$$\Rightarrow \quad y(t, T) = -\frac{\log P(t, T)}{(T - t)}.$$

The $y(t, T)$'s are referred to as spot rates, or zero rates.

Zero prices and the instantaneous interest rate

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- The instantaneous interest rate r_t is the interest rate on the MMA:

$$dB_t = r_t B_t dt, \quad \text{or} \quad B_t = B_0\, e^{\int_0^t r_u du}.$$

- If r_t is a constant r, the yield-to-maturity of any zero is r:

$$y(t,T) = r, \quad \forall\, t, T, \quad \text{and} \quad P(t,T) = e^{-r(T-t)}, \quad \forall\, T$$

- If r_t follows some stochastic process, the FTAP tells us that

$$P(t,T) = \mathbb{E}_t^Q \left[e^{-\int_t^T r_u du} \right].$$

- Since $P(t,T) = e^{-y(t,T)(T-t)}$, we can write

$$y(t,T) = -\frac{1}{T-t} \log \left[\mathbb{E}_t^Q \left(e^{-\int_t^T r_u du} \right) \right].$$

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Compounding frequency

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- Unless mentioned otherwise, time is measured in years and rates are annual rates. But the compounding frequency may vary.
- For short maturities, discrete infra-annual compounding is often used.
- For example, we can quote the interest rate for 6-month maturity at time t as an annual rate with semi-annual compounding, denoted $L(t, t+\frac{1}{2})$. It is such that

$$P\left(t, t+\frac{1}{2}\right) = \frac{1}{1+\frac{L(t,t+\frac{1}{2})}{2}}.$$

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"Simple" rates

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- More generally, we define the simple spot rate $L(t, T)$ by

$$P(t, T) = \frac{1}{1 + (T - t)L(t, T)}.$$

- \$1 invested at t grows to $1 + (T - t)L(t, T)$ at T.
- Typically $T - t = 1/12, 1/6, 1/4, 1/2$.
- For example, for $T - t = 1/12$, $L(t, T)$ is an annual rate compounded monthly.

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Annual compounding

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- For maturities longer than one year, it is frequent to express rates as annual rates with annual compounding.
- For an integer $n \geq 1$, we would write

$$P(t, t+n) = \frac{1}{[1 + y(t, t+n)]^n}.$$

- \$1 invested at t grows to $\$[1 + y(t, t+n)]^n$ at $t + n$

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Compounding frequency (summary)

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- When a per annum rate is quoted, the compounding frequency needs to be specified - continuous, infra-annual, annual.
- For example, at the per annum rate r, \$1 grows to ...
  - with continuous compounding, e^{2r} in 2 years
  - with continuous compounding, $e^{\frac{r}{2}}$ in 6 months
  - with annual compounding, $(1+r)^2$ in 2 years
  - with annual compounding, $(1+r)^{\frac{1}{2}}$ in 6 months
  - with semiannual compounding, $(1+\frac{r}{2})^4$ in 2 years
  - with semiannual compounding, $1+\frac{r}{2}$ in 6 months
  - with semiannual compounding, $(1+\frac{r}{2})^{\frac{1}{6}}$ in 1 month
- Henceforth, y will denote a continuously compounded yield-to-maturity, unless specified otherwise.

The yield curve

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- The yield curve (or zero curve) at time t represents the mapping $T \mapsto y(t, T)$
  - Also referred to as the term structure of interest rates.
- At the short end of the curve, we observe various zero rates directly, e.g. overnight LIBOR, 3-month LIBOR, etc.
- Zero rates for other maturities typically need to be extracted from the prices of coupon-bearing bonds.
  - Procedure known as bootstrapping (see below).

Bond pricing (given the yield curve)

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- Zero rates (or the prices of zero-coupons) are the DNA of fixed-income pricing (analogous to state prices).
- Zero rates are used to discount deterministic cash flows. Cash flows at different maturities are not discounted using the same discount rate: A safe cash flow at T_i is discounted back to t using $y(t, T_i)$
- A coupon bond is just a bundle of zero-coupons. Given the term structure of zero rates, absence of arbitrage requires that the price at time t of a bond paying a series of n cash flows c_i at dates T_i is

$$\Lambda_t = \sum_{i=1}^{n} e^{-y(t,T_i)(T_i-t)} c_i = \sum_{i=1}^{n} P(t, T_i) c_i$$

Note: $P(t, T_i)$ can be used directly as a discount factor.[1]

[1] Same thing for compounding: If you invest (resp. borrow) 1 at t with maturity date T_i, you get (resp. owe) $e^{y(t,T_i)(T_i-t)} = 1/P(t, T_i)$ at maturity.

Note on bootstrapping

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- The logic of bootstrapping is exactly the opposite - obtain the yield curve (or equivalently the prices of zero-coupons) from the prices of coupon bonds.
- Suppose you observe the prices of K bonds, Λ_t^k, for $k = 1, \ldots, K$
- Suppose, as of time t, bond k still has to pay $n(k)$ known cash flows. -Let c_i^k denote the cash flow on the bond at time $T_i^k \geq t$, for $i = 1, \ldots, n(k)$
- The problem is to find a set of zero-coupon bond prices such that all K bonds are priced consistently, i.e.

$$\Lambda_t^k = \sum_{i=1}^{n(k)} P(t, T_i^k) c_i^k, \quad \forall k = 1, \ldots, K.$$

Remark on the volatility of bond prices

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- Even though their cash flows are known, bonds are not boring; if the yield curve moves, their prices are affected.
- Example: Consider a zero-coupon bond that matures in 3 years. Its value in one year depends on the 2-year rate prevailing in one year.
- Bonds would be boring if the instantaneous interest rate r_t was a constant r
  -Then the yield curve would be flat at r: $y(t, T) = r$ for all t, T.
  - Zero-coupon prices would be deterministic: $P(t, T) = e^{-r(T-t)}$ for all t, T

Yield-to-maturity of a bond

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- Consider a 10-year maturity bond with a principal (or face value) of $ 100 and a coupon rate of 8% per annum, with semi-annual coupon payment.
- The cash-flows associated with this bond are: Twenty coupon payments of $\frac{0.08}{2} \times 100 = \$4$ every 6 months (starting 6 months from now) and a principal repayment of $ 100 at maturity (10 years from now).
  -Note that the coupon rate is quoted with the same compounding frequency as the payment frequency of the bond, semi-annual in this case.
- Taking the price Λ of the bond as given, its yield-to-maturity y (expressed as an annual rate with continuous compounding) is implicitly defined as

$$\Lambda = 4 \sum_{i=1}^{20} e^{-\frac{i}{2}y} + 100 e^{-10y}.$$

Yield-to-maturity (cont'd)

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- More generally, for a bond with cash flows c_i of tenor(time remaining until the cash flow is received) $\tau_i := T_i - t$, and given price Λ, the yield-to-maturity of the bond is defined implicitly by

$$\Lambda = \sum_{i=1}^{n} e^{-y\tau_i} c_i.$$

- Interpretation: The yield-to-maturity is that constant discount rate which, if applied to value the bond cash flows, gives the observed price of the bond.

Remarks on yield-to-maturity

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- The yield-to-maturity of a coupon bond has to be found by trial-and-error.
- For a given series of cash flows, yield-to-maturity and price convey equivalent information; you can get one from the other.
- The yield-to-maturity of a bond is typically not the rate of return on the bond investment - even if held until maturity![2]

[2] Except for zero-coupon bonds. For coupon bonds, the issue is at what rate you will be able to reinvest the coupon payments in the future.

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Forward rates

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- Let $F(t, T, S)$ denote the time t forward rate for future time period $[T, S]$.
- Using continuous compounding, this rate is defined by:

$$e^{y(t,T)(T-t)}e^{F(t,T,S)(S-T)} = e^{y(t,S)(S-t)}.$$

- This is a definition (not a no-arbitrage relationship). It follows that:

$$e^{F(t,T,S)(S-T)} = \frac{e^{-y(t,T)(T-t)}}{e^{-y(t,S)(S-t)}} = \frac{P(t,T)}{P(t,S)}.$$

Forward rates (cont'd)

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- Examples of forward rates: One-year rate in one year, two-year rate in one year, one-year rate in two years, etc.
- We can construct a forward curve. For instance,

$$\tau \mapsto F(t, t+1, t+1+\tau),$$

  i.e. the term structure of forward rates starting one year from now implied by today's spot rates.
- A different sort of curve would be

$$n \mapsto F(t, t+n, t+n+1),$$

  representing the evolution of the one-year forward rate implied by today's spot rates.

Relationship between current spot rates and forward rates

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- Recall the definition of $F(t, T, S)$

$$e^{y(t,T)(T-t)}e^{F(t,T,S)(S-T)} = e^{y(t,S)(S-t)}.$$

- Taking logs, we obtain:

$$y(t, S) = \frac{T-t}{S-t}y(t, T) + \frac{S-T}{S-t}F(t, T, S).$$

- Interpretation: A long-maturity spot rate is a weighted average of a short-maturity spot rate and the intervening forward rate.

Locking in the forward rate on future saving/borrowing

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- Saving or borrowing at the forward rate $F(t,T,S)$ for the period $[T,S]$ can be locked in at t by trading in zero-coupon bonds.
- Saving: Sell the T -bond, and invest the proceeds in the S - bond:

Position	t	T	S
Sell 1 unit of the T-bond	$+P(t,T)$	-1	0
Buy $\frac{P(t,T)}{P(t,S)}$ units of the S-bond	$-P(t,T)$	0	$\frac{P(t,T)}{P(t,S)}$
Net	0	-1	$\frac{P(t,T)}{P(t,S)}$

- Recall that
$$\frac{P(t,T)}{P(t,S)} = e^{F(t,T,S)(S-T)}.$$
- The zero-value bond portfolio at time t allows you to save 1 between T and S at the rate $F(t,T,S)$

Borrowing at the forward rate

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- Borrowing: Buy the T -bond, financing the purchase by selling S -bonds:

Position	t	T	S
Buy 1 unit of the T-bond	$-P(t,T)$	1	0
Sell $\frac{P(t,T)}{P(t,S)}$ units of the S-bond	$+P(t,T)$	0	$-\frac{P(t,T)}{P(t,S)}$
Net	0	1	$-\frac{P(t,T)}{P(t,S)}$

- The zero-value bond portfolio at time t allows you to borrow 1 between T and S at the rate $F(t,T,S)$

Forward rates defined as simple rates

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- Recall that, for a forward rate $F(t, T, S)$ defined as a continuously compounded rate,

$$e^{F(t,T,S)(S-T)} = \frac{P(t, T)}{P(t, S)}.$$

- We can instead define the forward rate $F(t, T, S)$ as a simple rate by

$$1 + (S - T)F(t, T, S) = \frac{P(t, T)}{P(t, S)}.$$

Forward rate agreements (FRAs)

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- A forward rate agreement (FRA) is an over-the-counter (OTC) contract between parties in which the underlying is an interest rate. In an FRA, the long side, is obligated to pay a fixed-rate interest on a given amount in the future.
- The payoff of an FRA (traded over-the-counter) is contingent on a realized floating rate. Therefore, an FRA can be used to hedge interest rate risk..
- Consider an FRA based on a (simple) floating rate $L(T, S)$
  - e.g. 6-month LIBOR prevailing in a year ($T = t + 1$ $S - T = 1/2$)
  - By investing 1 at T at the market rate $L(T, S)$, you get $1 + (S - T)L(T, S)$ at S
- For notational convenience, let $L_T := L(T, S)$; note that this rate is observed at T

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FRA payoff

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- An FRA stipulates a fixed interest rate L_K and a notional amount N.
- The payoff of an FRA, received at time S, is

$$(S-T)(L_T-L_K)\times N \quad \text{(for a long position)},$$
$$(S-T)(L_K-L_T)\times N \quad \text{(for a short position)}.$$

- Draw a payoff diagram as function of L_T, for given L_K
- There is no transfer of cash at the inception of the contract.
- Being long in an FRA gives protection against a high floating rate: insurance for a borrower. Being short pays off when the floating rate is low.

Pricing/Valuation of FRAs

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- Given that there is no transfer of money at the inception of the contract, what fixed rate L_K should be stipulated in the contract?
- Entering an FRA at time t allows you to lock in at t the borrowing/savings rate L_K over the future period of time $[T, S]$
- Locking in the savings rate L_K:

Position	t	T	S
Go short in an FRA at time t	0	0	$(S-T)(L_K - L_T)$
Save 1 at time T at L_T	0	-1	$1 + (S-T)L_T$
Net	0	-1	$1 + (S-T)L_K$

- By going short in an FRA with face value 1 at time t, and saving 1 at the prevailing rate L_T at time T, you earn the interest rate L_K over the period $[T, S]$ no matter what happens to L_T.

Pricing of FRAs (cont'd)

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- Locking in the borrowing rate L_K

Position	t	T	S
Go long in an FRA at time t	0	0	$(S-T)(L_T-L_K)$
Borrow 1 at time T at L_T	0	1	$-[1+(S-T)L_T]$
Net	0	1	$-[1+(S-T)L_K]$

- We saw that, by taking a position at t in zero-coupon bonds maturing at dates T and S, you can lock in the forward rate $F(t,T,S)$ over the period $[T,S]$
- Therefore, absence of arbitrage requires that

$$L_K = F(t,T,S),$$

where $F(t,T,S)$ is quoted as a simple rate.

Marking-to-market an FRA

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- Consider a long position in an FRA entered into at time 0 at the fixed rate $L_{K,0} = F(0, T, S)$
- What is the value of this FRA at time t, $(0 \leq t < T)$?
- An FRA entered into at time t has fixed rate $L_{K,t} = F(t, T, S)$
- At time t, you can go short in the current FRA (which can be done at zero cost),thus capturing the difference between the current FRA strike and the old FRA strike at time S

$$\begin{aligned}&(S-T)(L_T - L_{K,0})N + (S-T)(L_{K,t} - L_T)N \\ &\quad = (S-T)(L_{K,t} - L_{K,0})N.\end{aligned}$$

- Hence the value V_t of the old FRA at time t is

$$\begin{aligned}V_t &= P(t,S)(S-T)(L_{K,t} - L_{K,0})N \\ &= P(t,S)(S-T)[F(t,T,S) - F(0,T,S)]N.\end{aligned}$$

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Early settlement vs settlement in arrears

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- Since L_T is observed at date T,the time S payoff from the FRA is known at $T < S$. A contract specifying the payoff

$$\frac{(S-T)(L_T - L_K)N}{1+(S-T)L_T}$$

at date T is equivalent to a contract specifying the payoff $(S-T)(L_T - L_K)N$ at S . The fair fixed rate L_K is the same for both contracts, i.e. $F(t,T,S)$ at time t

- In practice, an FRA is typically early-settled, but it is more straightforward to analyze the equivalent late-settled FRA (especially since swaps, which are bundles of FRAs, are typically late-settled).

Interest rate swaps

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- An interest rate swap is just a bundle of FRAs of different maturities with the same strike level (fixed leg).
  - Effectively, an interest rate swap is an agreement to exchange periodically a fixed payment against a floating payment which depends on a (randomly) floating interest rate.
- Example: Every six months during the next 2 years (starting in 6 months), you will pay a fixed amount $\frac{1}{2} \times 5\% \times 100{,}000 = \$2{,}500$ and you will receive $\frac{L}{2} \times 100{,}000$ where L is the 6-month LIBOR (expressed as a simple rate) observed 6 months earlier.
- Only the net transfer is executed. At each settlement date $T_i = i/2$, for $i = 1, \ldots, 4$ your payoff is
  $0.5 \times [L(T_i - 0.5, T_i) - 5\%] \times 100{,}000$ \$.

Swap example

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- Suppose the following 6-month rates are observed over the term of the swap:
  $L(0, 0.5) = 4.8\%, L(0.5, 1) = 5.2\%, L(1, 1.5) = 5.4, L(1.5, 2) = 4.4\%$
  - after 6 months, you pay \$ 100 (known at inception),
  - after 1 year, you receive \$ 100,
  - after a year and a half, you receive \$ 200,
  - after 2 years, you pay \$ 300.
- Terminology:
  -The fixed rate 5% is the swap rate,
  - \$ 100,000 is the notional,
  - You receive floating and pay fixed $\rightarrow$ you are long in the swap.
- There is no initial transfer of money, i.e. the swap rate is fair when the contract is initiated (we will see later how the fair swap rate is pinned down).

Using swaps

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- Interest rate swaps allow you to get rid of floating rate risk on your liabilities or assets.
- You go long in a swap if you want to transform a floating-rate loan into a fixed-rate loan. Then at each settlement date, you pay a fixed amount and you receive from your swap counterparty a variable amount corresponding to the interest due on the loan.
- You go short in a swap if you want to transform an asset earning a floating interest rate into an asset earning a fixed interest rate. Then at each settlement date, you use the asset's variable cash flows to pay the floating amount due on the swap and you receive from your swap counterparty a fixed interest rate.

Swaps: Notation

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- A swap entered into at time t specifies:
  - A swap rate s_t;
  - n payment dates $T_i = t + i\Delta$, for $i = 1, \ldots, n$;
  - A floating rate $L(T_{i-1}, T_i)$, for $i = 1, \ldots, n$;
  - A notional N.
- Δ is the swap tenor, the length of time between payment dates.
- T_n is the swap term or maturity.
- Both fixed and floating rates are quoted as simple rates (i.e. with compounding frequency equal to Δ).
- A long position in the swap generates a sequence of payoffs

$$\Delta[L(T_{i-1}, T_i) - s_t]N, \ i = 1, \ldots, n.$$

- In the preceding example:

$$t = 0, s_t = 5\%, n = 4, \Delta = 1/2, T_n = 2, N = 100,000$$

Determination of the fair swap rate: Approach #1

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- Definition: A floating-rate bond with tenor Δ and n payment dates pays a coupon at date $T_i = i\Delta$, $i = 1, \ldots, n$, at the rate of $L(T_{i-1}, T_i)$, plus the principal at maturity T_n
- Observe that the payments in a swap are exactly as if a floating-rate bond was exchanged against a fixed-rate bond, both bonds having face value equal to the notional of the swap (N)
  $\rightarrow$ going long in a swap (i.e. receiving floating, paying fixed) is like buying a floating-rate bond and selling a fixed-rate bond at the same time;
  $\rightarrow$ going short in a swap is like buying a fixed-rate bond and selling a floating-rate bond at the same time.
  Note that the principal repayments at maturity cancel out.
- Given the current zero curve, what is the fair swap rate on a swap? In order for the swap to be worth zero initially, the swap rate must be such that the value of the fixed-rate bond is equal to the value of the floating-rate bond.

Fair swap rate (cont'd)

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- The fixed-rate bond implicit in the swap pays a fixed coupon rate S_t, i.e. a coupon payment of $\Delta s_t N$ at each payment date $T_i, i = 1, \ldots n$, plus the principal N at maturity T_n. Its value given the yield curve at time t is

$$\sum_{i=1}^{n} P(t, T_i)\Delta s_t N + P(t, T_n)N$$

- The floating-rate bond implicit in the swap is worth its face value N
  Proof: The payments of the bond can be replicated by saving N at the current floating rate and subsequently, at each payment date, collecting the coupon and rolling over N at the floating rate at that time. The intial cost of this self-financing strategy is N.

Fair swap rate (cont'd)

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- Therefore, at time t, the swap rate s_t must be such that:

$$\sum_{i=1}^{n} P(t, T_i)\Delta s_t N + P(t, T_n)N = N.$$

- Solving for s_t, we get

$$s_t = \frac{1 - P(t, T_n)}{\Delta \sum_{i=1}^{n} P(t, T_i)}.$$

- Unsurprisingly, the swap rate is entirely determined by the yield curve at the initiation of the contract.
- By construction, the swap rate is equal to the par yield (i.e. the fixed coupon rate such that a bond paying coupons at this rate would be valued at par).

Determination of the fair swap rate: Approach #2

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- A swap is a bundle of FRAs, where the strike of each FRA is the swap rate.
- The swap payment at T_i is

$$\Delta[L(T_{i-1}, T_i) - s_t]N.$$

- By entering into a short position in the corresponding current (zero-value) FRA, we get a non-stochastic cash flow at T_i

$$\begin{aligned}&\Delta[L(T_{i-1}, T_i) - s_t]N + \Delta[F(t, T_{i-1}, T_i) - L(T_{i-1}, T_i)]N \\ &= \Delta[F(t, T_{i-1}, T_i) - s_t]N,\end{aligned}$$

- the value of which at time t (using the date t yield curve) is

$$P(t, T_i)\Delta[F(t, T_{i-1}, T_i) - s_t]N.$$

Fair swap rate (cont'd)

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- The fair swap rate is such that the sum of the values of the FRAs implicit in the swap is zero, i.e. s_t solves

$$\Delta N \sum_{i=1}^{n} P(t, T_i)[F(t, T_{i-1}, T_i) - s_t] = 0$$

$$\Rightarrow \quad s_t = \sum_{i=1}^{n} \frac{P(t, T_i)}{\sum_{k=1}^{n} P(t, T_k)} F(t, T_{i-1}, T_i). \tag{S2}$$

- The fair swap rate is a weighted average of forward rates.
- Remark: If the yield curve is upward sloping, for the value of the swap to be zero initially, it must be that, from the point of view of the long party, the early FRAs have negative value and the late FRAs have positive value.

Fair swap rate (cont'd)

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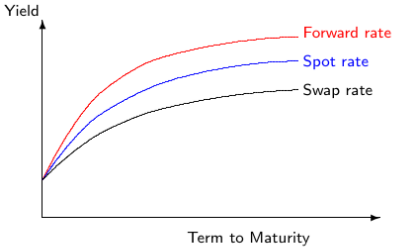
- Using the fact that

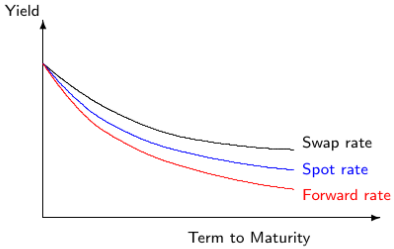
$$1 + \Delta F(t, T_{i-1}, T_i) = \frac{P(t, T_{i-1})}{P(t, T_i)},$$

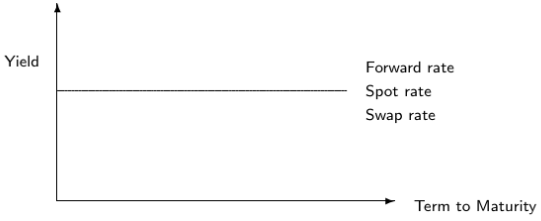
it is straightforward to check that the fair swap rate formula (S2) is the same as the formula (S1) that we derived using the first method.

Spot, forward and swap rates

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Marking-to-market of a swap

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- After a swap contract has been initiated it can have non-zero value.
- At a given point of time t, the value of a long swap position is equal to the value of the floating-rate bond minus the value of the fixed-rate bond.
- The value of the fixed-rate bond at date t is obtained directly by valuing known future payments using the date t yield curve.
- If the next payment date after t is t' and the (known) floating coupon at t' is c, the value of the floating-rate bond at t is $P(t,t')(c+N)$
  -Recall that the value of a floating-rate bond at initiation is N
  - The ex-coupon value of a floating-rate bond at any payment date is also N, since the remaining payments can be replicated by saving N at the floating rate and, at each subsequent payment date, collecting the coupon and rolling over N at floating rate at that time.

Marking-to-market of a swap (cont'd)

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- Alternatively, a swap can be valued as a bundle of FRAs.
  - The value of a long position in a swap, at a swap rate of s_0, and n remaining payments, is given by the present value of the first net payment of the swap, which is known (since it is contingent on a floating rate that has already been observed), plus

$$\Delta N \sum_{i=2}^{n} P(t, T_i)[F(t, T_{i-1}, T_i) - s_0].$$

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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